DS	**Session**	Queue	**Question Information**	Level 1 Challenge 60

Question description

Umesh is an DS expert training youngsters struggling in DS to make them better.

Umesh usually gives interesting problems to the youngsters to make them love the DS.

One such day Umesh provided to the youngsters to solve the task such that, Reverse a Queue, Queue data structures work on the FIFO architecture so the element that has entered first in the list will go out from the list first.

Youngsters were lacking the idea to solve the problem.

Being an exciting youngster can you solve it?

Function Description

- INITIALIZE FRONT AND REAR = -1
- CALL ENQUEUE FUNCTION WHILE(I<GIVEN SIZE OF QUEUE)
- CALL REVERSE
- MAKE A FOR LOOP(I=FRONT,J=REAR;I<J;I++;J++)
- SWAP(FRONT,REAR)

Constraints:

0<size <100

0<data<1000

Input format:

First line indicates the size of the queue

Second line indicates the elements of the queue.

Output Format:

first line indicates the enqueue of each elements

last line indicates the reversed queue elements.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
5 6 3 1 2
```

EXPECTED OUTPUT

```
Queue:5 6 3 1 2
Reversed Queue:2 1 3 6 5
```

Test Case 2

INPUT (STDIN)

```
8
1 8 6 4 9 5 3 5
```

EXPECTED OUTPUT

```
Queue:1 8 6 4 9 5 3 5
Reversed Queue:5 3 5 9 4 6 8 1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void enqueue(int data,int l)
```

Test Case 2

KEYWORD

```
void reverse()
```

Test Case 3

KEYWORD

```
enqueue(t,n);
```

Test Case 4

KEYWORD

```
reverse();
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
4
```

Test Case 2

TOKEN COUNT

```
275
```

Test Case 3

NLOC

```
57
```

Code

solution.c

```c
#include <stdio.h>

int queue[100];
int front = 0, rear = -1;

void enqueue(int data, int l) {
    queue[++rear] = data;
}

void reverse() {
    int i = front, j = rear;
    while (i < j) {
        int temp = queue[i];
        queue[i] = queue[j];
        queue[j] = temp;
        i++; j--;
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    for (int i = 0; i < n; i++) {
        int t; scanf("%d", &t);
        enqueue(t, n);
    }
    printf("Queue:");
    for (int i = front; i <= rear; i++) printf("%d ", queue[i]);
    printf("\n");
    reverse();
    printf("Reversed Queue:");
    for (int i = front; i <= rear; i++) printf("%d ", queue[i]);
    printf("\n");
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
5
5 6 3 1 2
Queue:5 6 3 1 2
Reversed Queue:2 1 3 6 5
PS C:\C Programming>
```